Deakin University Trimester 2, 2025
School of IT Assignment 1
Unit Team: SIT742

SIT742: Modern Data Science

Extension Request Students with difficulty in meeting the deadline because of various reasons, must apply for an assignment extension no later than 8:00pm on 04/08/2025 (Monday). Apply via '`CloudDeakin`', the menu item '`Extension Request`' under the '`Assessment`' drop-down menu.

Academic Integrity All assignment will be checked for plagiarism, and any academic misconduct will be reported to unit chair and university.

Generative AI Deakin's Policy and advices on responsible usage of Generative AI in your studies: `https://www.deakin.edu.au/students/study-support/study-resources/artificial-intelligence`

Instructions

Assignment Questions

There are total **2** main parts and **1** optional part in the assessment task:

Part 1 The first part will focus on the basic Python programming skills which includes the `data types`, the `control flow`, the `function`, the `modules and library` from **M02**.

Part 2 The second part is for those who are aiming to achieve '`High Distinction`' (HD) for this assessment task, and it will focus on more advanced Python programming skills for data science on particular scenario. This part will require the knowledge covered in **M02** and also some part of the **M03**, in particular, `numpy` and beyond.

Part 3 **(Optional)** This part is **optional** and carries **0** mark.

What to Submit?

You are required to submit the following completed files to the corresponding *Assignment* (Dropbox) in CloudDeakin:

`SIT742Task1.ipynb` The completed notebook with all the run-able code on all requirements.

In general, you need to complete, save the results of running, and submit your **notebook** from Python platform such as `Google Colab`. You need to clearly list the answer for each question, and the expected format from your notebook will be like in Figure 1.

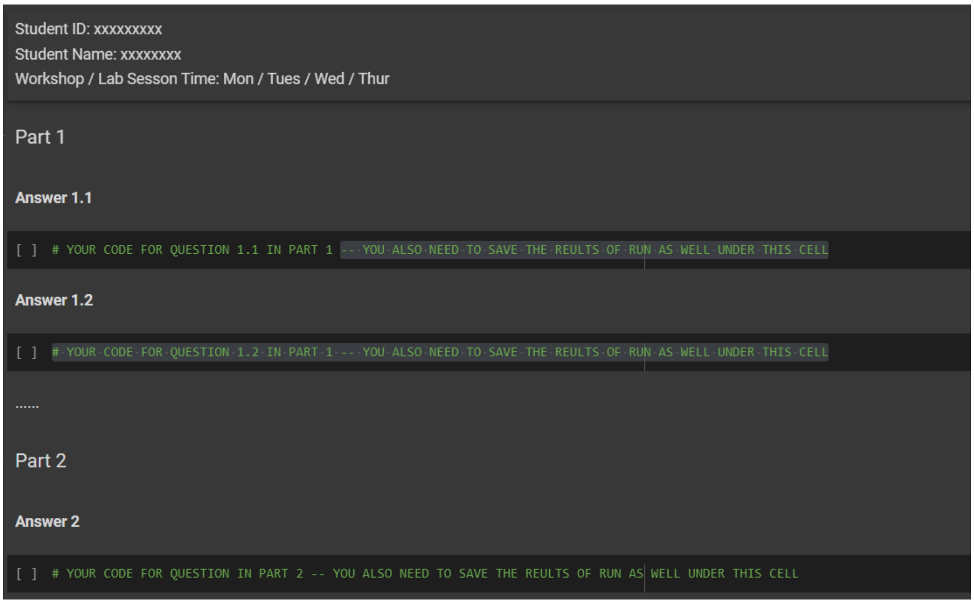

Figure 1: Notebook Format

`SIT742Task1Part2.avi` (Only if you choose to work on *Part 2* of this assessment task) A video demonstration within 5 minutes, and the file format can be other common ones, such as '`MKV`', '`WMV`', '`MOV`' etc.

If you are aiming to achieve '`High Distinction`' (HD) and choose to work on *Part 2* of this assessment task, one important submission is a **short video** in which *You* orally present the understanding of the solutions / scenarios that you provide in the notebook and illustrate the running of code line by line.

`SIT742Task1Part2Report.pdf` (Only if you choose to work on *Part 2* of this assessment task) A short essay report with no more than 800 words, and the file format can be other common ones, such as '`.pdf`', '`.doc`', '`docx`' etc.

If you are aiming to achieve '`High Distinction`' (HD) and choose to work on *Part 2* of this assessment task, another important submission is a **short essay** in which *You* will explain the the solution with given requirements, any of the resource or material which has been helpful to resolve the problem, and illustrate if there is any alternative way to solve the problem with your own critical review with given requirements. In addition, you will also need to follow the common essay writing structure which could be found in `https://www.deakin.edu.au/students/study-support/study-resources/academic-skills/essay-writing`

Part I

Python Programming

There are **7** questions in this part for total **70** marks, and each question is for **10** marks.

You are required to use `Google Colab` to finish all the coding in the *code block cell*, and provide sufficient coding comments, and also save the result of running as well.

Question 1.1

Write a function, with **string** input of your **student id** starting with "s". The function will take all the characters in your student id and then reverse the order into list. *For example, if you input "s123456", then the function output will be "['6','5','4','3','2','1','s']".*

Please consider handling exceptions and list comprehension if necessary.

Please also write down your thoughts on the code optimization with **comments** as well.

Question 1.2

Continue modifying the function you defined in Question 1.1 by adding an additional **numerical** integer input, which specifies the index from which to start reversing the student ID string, *including the character at that index.*
For instance, given the input "s123456" and 3, the function should return "s126543".
Ensure to handle possible exceptions appropriately, and consider using list comprehension if applicable.

Additionally, include your thoughts on potential code optimization using **comments** within your implementation.

Question 1.3

Write a code which uses your student ID **integer digits** as input, then transforms it into a list and prints the maximum number and the index location (if it is asked) with the below requirements.

Please do not use default functions such as `max()`, `sort()` in this task.

- Write a code by using list comprehension that prints the maximum number and the index location.
- Write a code to find out the second maximum number and if the second maximum does not exist, return the maximum number.

To make it simple, please only consider the distinct numbers.

Question 1.4

Write a function "`find_digits`" that takes your student id **integer digits** as input and resolves the problem of "how many numbers are smaller than the current number for each digit". *For example, if you input "123456", then for 1, there are 0 numbers smaller than it; for 2, there are 1 number smaller than it (1); for 3, there are 2 numbers smaller than it (1,2).*

Be sure to test your function well; also, please write down your thoughts on the code optimization with **comments** as well, particularly please discuss your **own thoughts** of **time complexity**.

Question 1.5

Deakin has a "*Three Minutes Thesis*" (`3MT`) competition day happening every year in which the higher degree research students compete in the presentation of their thesis. A group of 5 students wanted to determine who could win the competition of day based on the three tasks results – the research citations (higher the better), the comprehension of the presentation (higher the better) and the typographical mistakes during presentation (lower the better). The table below shows their results of the three tasks and each task has the equal importance. Please **design a function** to determine who could win the competition with **your own reasonable logic**. Please write down the reason why you choose that student and also your logic of your code as well.
`Hint: please consider the data wrangling with data processing, and justify the processing technologies you are using.`

Student Name	Research Citations	Presentation Comprehension	Typographical Mistake
Kelvin	520	6	0.11
Lily	27	7	0.05
Thomas	120	3	0.16
Dina	310	5	0.08
Andrew	55	8	0.01

Question 1.6

Gaming Company `Deakin Magic` invites you to design a code for one of the scenarios for game character Allan in Game "Magic Land". In the game, Allan will need to get magic coins with two magic machines (the machine will consume some coins and generate some for Allan) shown as below table:

Magic Machines	Used Coins	Coin Generated
Machine 1	x	2x+1
Machine 2	x	2x+2

Your code needs to work out the solution of the way on how to use the two machines to generate `n` total coins for Allan.
Please write down **your own understanding** of the problem and also the explanation of the solution in comments. Also run your code with 3 different total coins to demonstrate your code effectiveness. `Hint:x could be 0`.

Question 1.7

Word Count Problem Create a string variable by your own with few sentences together and finish below requirements:

- Write a function to split the text into sentences (using full stops, exclamation mark or question mark). Each sentence should be stored as a separate string in a list. For example, given string text "how are you? i am good.", your sentence list should be "['how are you?','i am good.']"
- Use the sentence list to find out the most common used word in the whole document and return the count of the top 30 most common words with dictionary type (sorted in descending order of frequency.). Please do not use `NLTK` or `Gensim` to perform this task (you do not need to import any library).

Part II

Python Foundations for Data Science

This part is for students who are aiming to achieve '`High Distinction`' (HD) for this assessment task. The Part II has been designed for a particular scenario based data science problems.

There are **2** versions in this part for **30** marks: **15** marks for coding, **5** marks for video presentation `SIT742Task1Part2.avi` (as in '**What to Submit**') and **10** marks for the essay `SIT742Task1Part2Report.pdf` (as in '**What to Submit**').

For your version of this part, you are required to use `Google Colab` to finish all the coding in the *code block cell*, provide sufficient coding comments, and also save the result of running.

1 Which question for you?

```
def sum_digits(n):
    r=0
    while n:
        r, n = r + n % 10, n // 10
    return r

def check_studentid(studentid):
    x = sum_digits(studentid)
    if x % 2 == 0:
        print ('version I')
    else :
        print ('version II')

print('Correct Version of Q2 for me is:')
#check_studentid ()
```

2 Question 2 - Version-I

Question 2: A Linear Optimization Problem with Numpy

Owning the grocery store will need to find out the bundle sale with maximum sales. Assuming for product A and B, a bundle sale has been designed, and Product A is \$3 per unit and product B is \$4. With the current logistic and transportation, there are some limitations on how to ship the two products together via airplane. Particularly, the units on product A and B should follow two general equations as below:

$$\begin{aligned} \texttt{Maximum} \quad & 3A + 4B \\ \texttt{Subject to:} \quad & A + 2B \leq 14 \\ & B \geq 3 \\ & A < 15 \\ & B < 15 \end{aligned}$$

Please follow the below questions to finish the optimization problem to find the optimal units for Product A and B respectively.

Question 2.1
The first step is to finish up the code for resolve above problem. However, there are some errors as well in the given code, could you please fix and continue on your code?

```
import numpy as np

# Define range to search
x_vals = np.arange(0, 15, 1)
y_vals = np.arange(0, 15, 1)

# Coefficients for objective
c = np.array([3, 4])
```

```
    # Storage for maximum solution
    max_val = -np.inf
    max_sol = (0, 0)
```

Question 2.2

The second step is to understand how linear optimization could help us in coding practice. In our IT-related environment, linear optimization could involve scenarios like `Resource Allocation`, `Package Dependency Resolution` and `Network Routing`. Could you please further write down your understanding with concrete examples on one scenario in the essay and also your video presentation? The essay will need to cover the details of the scenarios, the problems definition, existing solution on linear optimization and your critical review or evaluation of the using linear optimization.

3 Question 2 - Version-II

Question 2 Stochastic Optimization with Numpy

For grocery store, you, the manager are now trying to find out the pattern between profit and stock level for each day. For product A and B, the demands of sales could be followed with normal distribution and you need to find out the optimal inventory stock of each products at day. If your stock is low, then you miss out on potential profit and if your stock is too high, you incur holding cost on unsold items. The goal here is to find the optimal number of units to stock that maximizes expected profit, considering random demand at day level, also analysis on the relationships between `Expected Profit` and `Stock Level` at day level.

The product information are given as below:

Product_Name	Selling price	Stock cost*	Holding cost*	Stock Range
Product A	20	8	2	20 - 80
Product B	15	7	1.5	15 - 70

* for the stock cost, it means the price to pay to the wholesalers and the total cost of the product is: cost total = stock * stock cost + unsold * holding_cost.

Question 2.1

In here, let's firstly finish the code for a `Monte Carlo Simulation` for this optimization problem here with product A only. You need to define some variables **by yourself** and provide reasonable justifications in the comments of your variable definitions.

```
import numpy as np
import matplotlib.pyplot as plt

# ---------------------------------
# Product Information goes here,
# ---------------------------------
selling_price = 20
cost_price = 8
holding_cost = 2

mean_demand =  # Define your own mean value of the daily demand
std_demand =  # Define your own std value of the daily demand

stock_range =    # Add stock defination here
```

```
n_simulations =     # Number of Monte Carlo simulations

# ------------------------------------------------
# Monte Carlo Simulation -- run with n_simulations
# ------------------------------------------------
expected_profits = []

for stock in stock_range:
    profits = []
    for _ in range(n_simulations):
    # continue your code below
```

Question 2.2

Now, let's move to multiple products with budget constraints as well. Please continue modifying the Question 2.1's code and run the optimization with both Product A and B with your own variable definitions. The budget is $2000 for all the stocking costs for the day.

Question 2.3

Please conduct the analysis between the stock level and profit with the example in Question 2.2. Also in our IT related environment, the stochastic optimization could be used in many scenarios, please write an essay regarding the particular IT related scenario and brief your proposed scenario in video? The essay will need to cover the details of the scenarios, the problem definition, existing solution on stochastic optimization and your critical review or evaluation of the using stochastic optimization.

Part III

Optional: Questionnaire on *Integrating Indigenous Perspectives into DS Education*

Note: This part of the assignment is **optional** and carries **no marks**.

In modern data-driven environments, collaboration often involves individuals from diverse cultural and social backgrounds. As part of our ongoing efforts to develop inclusive and socially responsible unit, we are exploring ways to meaningfully integrate Australian Indigenous perspectives into Data Science education, specifically in the context of the unit *SIT742: Modern Data Science*. We invite you to share your thoughts and feedback on this important topic. Your input will help us improve the quality and inclusiveness of both the unit and its associated learning materials.

If you are willing to assist, please consider completing a short questionnaire titled *Integrating Indigenous Perspectives into Data Science Education*, which can be accessed at the following link:

- Pre-Survey Questionnaire

(Please interpret its references to "`ICT`" as also including *Data Science*.)

Related Information

Human Research Ethics Application ID: `2025/HE000518`

Approval Date: `28/04/2025`

Estimated Time to Complete: Approximately 10 minutes

Confidentiality: All responses are strictly confidential and will be used solely for research and educational improvement purposes.

University Policy Deakin Indigenous Strategy 2023-2028